
Digantara Company Assessment

Question 2 — Technical Response

Image Preprocessing • Tiling • Pixel-Level Segmentation Methodology

2a. Image Pre-processing Steps

The preprocessing pipeline was designed to enhance the visibility of faint astronomical features (stars/blobs and streaks/objects) while preserving scientific integrity and minimizing noise amplification. The following steps were applied systematically and identically to all 10 FITS images provided in the dataset.

Step 1 — Data Type Conversion and Normalization

Raw FITS data was read exclusively from the primary Header Data Unit (HDU) and immediately converted to 32-bit floating-point format. This conversion provides the numerical headroom required for precise percentile calculations and stretch operations without intermediate truncation or overflow artifacts.

Step 2 — Percentile Stretch (Contrast Enhancement)

A robust percentile stretch using the **0.5th** and **99.8th** percentiles was applied as the lower and upper bounds respectively. This approach is preferred over simple min-max normalization for astronomical imagery because it:

- Mitigates the influence of cosmic rays and hot-pixel outliers that would otherwise dominate the dynamic range.
- Simultaneously renders both faint stars and bright streaks visible within a single display range.
- Preserves the relative intensity relationships between genuine celestial features across the field.

$$I_{0.5} = \text{clip} \left(\frac{(I - P_{0.5})}{(P_{99.8} - P_{0.5} + \epsilon)}, 0, 1 \right) \text{ where } \epsilon = 10^{-8} \text{ normalized raw}$$

Step 3 — Median Filtering (Noise Reduction)

A 3×3 median filter was applied after the stretch had been quantized to 8-bit. Median filtering is particularly suitable for this data because it preserves sharp edges critical for streak delineation, is robust to residual salt-and-pepper noise, and maintains the approximately circular morphology of unresolved star blobs.

Step 4 — Quantization and Format Conversion

The stretched floating-point data were converted to 8-bit unsigned integers (0–255) and written as lossless PNG files. PNG was selected for its universal compatibility with annotation platforms (CVAT, Roboflow, Label Studio). Quantization error remains below 0.4 % per pixel — negligible for subsequent visual annotation.

Quality Assurance

After preprocessing, every image was visually inspected. Confirmation criteria included: clear visibility of faint stars (local contrast $\geq 0.1\%$ above background), distinguishable linear streaks (contrast $\geq 0.05\%$ above the noise floor), and an acceptable signal-to-noise ratio in empty sky regions.

2b. Boundary Handling and Tiling Strategy

Each preprocessed image measures 9568×6380 pixels. The required tile size is exactly 1024×1024 pixels. Because neither dimension is an integer multiple of 1024, a deterministic zero-padding strategy was adopted to guarantee that **no original pixel is discarded**.

Dimensional Analysis

Axis	Original Size	Full Tiles	Remainder	Padding Added
Width	9568 px	9	368 px	656 px (right edge)
Height	6380 px	6	236 px	788 px (bottom edge)

After padding the image becomes **$10\,224 \times 7\,168$** pixels, which divides exactly into a 10×7 grid (70 tiles per image). All padding pixels are set to intensity 0 (black).

Implementation Sequence

1. Load the preprocessed 9568×6380 image into memory.
2. Apply zero-padding with OpenCV's `copyMakeBorder(..., borderType=BORDER_CONSTANT, value=0)`.
3. Extract successive non-overlapping 1024×1024 tiles in row-major order.
4. Write a companion JSON metadata file recording original dimensions, padding amounts, and the (row,column) origin of every tile.

Rationale for Zero-Padding

Several alternative strategies (overlapping tiles, cropping, or reflective padding) were considered. Zero-padding was selected for the following reasons:

- **No data loss** — every original pixel is retained; padding is purely structural.
- **Semantic validity** — intensity 0 corresponds to the absence of signal, consistent with empty sky.
- **Annotation clarity** — human annotators can immediately recognize and ignore padded regions.
- **Perfect repatching** — after annotation the padded borders are simply cropped away, restoring the exact original geometry.

The stored JSON metadata guarantees that any subsequent full-frame reconstruction is lossless and spatially accurate, satisfying the requirement to submit repatched full-size imagery for visual inspection.

2c. Method and Criteria for Distinguishing Blobs and Streaks

Accurate class assignment is essential for generating reliable pixel-level masks. The distinction between the two classes rests on quantitative morphological measurements computed for every connected component after an initial intensity threshold.

Blob / Star (Class 0)	Streak / Object (Class 1)
Compact, roughly circular point sources corresponding to unresolved stars. • Aspect ratio: $0.7 \leq AR \leq 1.4$ • Eccentricity: $e \leq 0.6$ • Area: typically 20–500 px • Simply connected (no holes)	Linear or gently curved features from satellites, debris or cosmic rays. • Aspect ratio: $AR > 1.8$ • Thin-line morphology • Length: 50 – 2000+ px • Arbitrary orientation

Operational Decision Flow

Step 1 — Compute the aspect ratio AR of the connected component.

- If $AR > 1.8 \rightarrow$ assign **Class 1 (Streak)**.
- Otherwise proceed to Step 2.

Step 2 — For components with $AR \leq 1.8$ evaluate eccentricity and area:

- If $e \leq 0.6$ and $area \in [20, 500]$ px $\rightarrow$ assign **Class 0 (Blob)**.
- All remaining ambiguous cases are subjected to manual visual review.

This two-stage rule set cleanly separates the large majority of features while leaving only a small number of short/fat borderline objects for human arbitration, thereby combining algorithmic consistency with expert oversight.

2d. Criteria for Annotating Faint or Small Blobs

Astronomical frames routinely contain point sources whose peak intensity is only marginally above the local background noise. These faint stars remain scientifically relevant and must be annotated consistently. The following dual criteria were applied:

Criterion 1 — Local Contrast Threshold

The peak intensity of a candidate blob must exceed the mean of a surrounding 50×50 pixel background window by at least **three standard deviations (3σ)**. Under the assumption of approximately Gaussian noise this threshold corresponds to a $\sim 99.7\%$ confidence that the detection is not a random fluctuation.

Criterion 2 — Morphological Circularity

In addition to the contrast requirement the candidate must exhibit a shape consistent with an unresolved point source:

- Circularity = $4\pi \times \text{Area} / \text{Perimeter}^2 \geq 0.7$ (a perfect circle yields 1.0).
- Aspect ratio remains inside the interval $[0.7, 1.4]$ already defined for the blob class.

Only sources satisfying **both** criteria were labelled as the star/blob class. The practical annotation workflow consisted of: (i) visual confirmation that the stretch had rendered faint sources visible, (ii) optional proposal generation with a segmentation model, (iii) strict verification against the two numeric thresholds, and (iv) uniform application of the same rules across every tile of every image.

Summary of Key Assumptions

- Only the primary HDU of each FITS file was used (single-band grayscale data).
- All geometric operations were performed in raw pixel coordinates; no WCS transformation was applied.
- No specialised cosmic-ray rejection or bad-pixel masking algorithms were employed.
- Zero-padding is accepted as a valid and reversible tiling strategy.
- Background noise is treated as approximately Gaussian when computing local σ thresholds.

The preprocessing, tiling and annotation methodology described above yields reproducible, high-quality pixel-level segmentation masks that are suitable for training modern computer-vision models. The systematic treatment of image boundaries, the quantitative morphological classification rules, and the explicit criteria for faint-source inclusion together constitute a complete and auditable pipeline consistent with best practices in astronomical image analysis.